INCH-POUND

MIL-M-14/7
2 April 1990

MILITARY SPECIFICATION SHEET

MOLDING COMPOUNDS, SILICONE, THERMOSETTING, CONTAINING MINERAL/GLASS FILLERS

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-M-14.

Silicone resin molding compounds, shall be of the following types, as specified:

Type MSG	–	Mineral filler, general purpose
Type MSI-30	–	Glass fiber, filler, impact-resistant; nominal-impact strength, 3.0 foot-pounds per inch notch.

REQUIREMENTS

Qualification test requirements are specified in table I. Batch acceptant test requirements are specified in table II.

NOTES

Type MSG. This type is a mineral-filled silicone compound of low-loss, high-dielectric strength, and excellent heat resistance.

Type MSI-30. This type is a glass-fiber-filled silicone compound of high-impact strength and heat resistance, but somewhat poorer electrical properties than type MSG.

AMSC N/A FSC 9330

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TABLE I. *Qualification test requirements for silicone resin molding compounds, mineral/glass filled.*

Requirement	Type MSG	Type MSG-30
	Mechanical/physical	
Compressive strength, endwise	15,000	10,000
Flexural strength	6,000	7,000
Heat deflection temperature	200	200
Heat resistance	200	200
Impact strength, side[1]	0.25	3.2
Tensile strength	2,500	2,000
Water absorption	0.50	0.50
	Electrical	
Arc resistance	210	175
Comparative track index	–	–
Dielectric breakdown:	Voltage in step-by-step test	
Short-time test	(To be recorded as basis for initial)	
Step-by-step-test	35	30
Short-time test	(To be recorded as basis for initial)	
Step-by-step test	35	30
Dielectric constant:		
at 1 kilohertz	5.0	5.0
	5.2	5.5
at 1 megahertz	4.7	4.7
	5.0	5.1
Dielectric strength:		
Short-time test	325	160
Step-by-step test	300	125
Short-time test	300	75
Step-by-step test	275	50
Dissipation factor:		
at 1 kilohertz	0.015	0.015
	0.020	0.050
at 1 megahertz	0.010	0.010
	0.015	0.060
Surface resistance	1,000	–
Volume resistance	1,000	–

See footnote at end of table.

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TABLE I. *Qualification test requirements for milicone resin molding compounds, mineral/glass filled – Continued.*

Requirement	Type MSG	Type MSG-30
Combustion		
Flame resistance:		
Ignition time	90	90
Burning time	120	120
Flammability	V-1	V-1
Toxicity when heated:		
Carbon dioxide	15,000	15,000
Carbon monoxide	1,000	1,000
Ammonia	2,500	2,500
Aldehydes as HCHO	50	50
Cyanide as HCN	60	60
Oxide of nitrogen at NO_2	100	100
Hydrogen chloride	100	100

[1]The side of the test specimen is that area formed by the chase of the mold.

TABLE II. *Batch acceptance test requirements for silicone resin molding compounds, mineral/glass filled.*

Property to be tested	Type MSG	Type MSI-30
Arc resistance	210	175
Dielectric constant at 1 megahertz	5.0	5.1
Dielectric strength, step-by-step	275	275
Dissipation factor at 1 megahertz	0.015	0.060
Flexural strength	6,000	7,000
Impact strength, side[1]	0.25	3.2
Water absorption	0.50	0.50

[1]The side of the test specimen is that area formed by the chase of the mold.

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